

SYNOPSIS

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QR codes (Quick Response codes) have become an integral part of digital communication, allowing users to store and retrieve information quickly using QR scanners or smartphone cameras. They are widely utilized in industries such as marketing, authentication, event ticketing, and contactless payments due to their ease of use and efficiency. This project aims to generate QR codes using Python, showcasing their potential in enhancing digital accessibility and information sharing.

The primary objective of this project is to create a QR code that encodes the URL of the PPG College website. This QR code can be scanned using any mobile device or QR scanner to instantly access the college's website, providing a seamless and efficient way of sharing information. The implementation is carried out using Python's `qrcode` library, which facilitates the creation and customization of QR codes. The generated QR code is displayed in ASCII format within the console for visual representation and is also saved as an image file for easy distribution and use in digital and print media.

This project highlights the significance of QR codes in modern technology and demonstrates their practical applications in different domains. By leveraging Python programming, it showcases a simple yet powerful method to encode and distribute information. The project serves as a foundation for further exploration of QR code applications, such as secure authentication, product tracking, and digital payments, illustrating how technology can bridge the gap between physical and digital interactions.

INTRODUCTION

1. INTRODUCTION

1.1ABOUT THE PROJECT

In today's digital age, QR codes (Quick Response codes) have become an essential tool for storing and sharing information efficiently. Originally developed for the automotive industry, QR codes have now found widespread applications in marketing, authentication, digital transactions, event management, and web accessibility. Their ability to store various types of data, such as URLs, text, and contact information, makes them a highly versatile solution for quick and seamless information retrieval.

This project focuses on generating a customized QR code using Python, specifically designed to provide instant access to the PPG College website. The implementation utilizes Python's qrcode library to generate and customize the QR code. The generated QR code is displayed in ASCII format within the console and is also saved as an image file, ensuring easy distribution and accessibility. Users can scan the QR code using their mobile devices to be redirected to the college website instantly.

1.2 SYSTEM SPECIFICATION

To successfully implement the QR code generation project using Python, the following hardware and software requirements must be met.

1.2.1 HARDWARE SPECIFICATION

For smooth execution and performance, the project requires a system with the following minimum and recommended hardware specifications:

- **Processor:**

- Minimum: Intel Core i3 (or equivalent AMD processor).
- Recommended: Intel Core i5/i7 (or higher) for better processing speed.

- **RAM:**

- Minimum: 4GB (sufficient for basic execution).
- Recommended: 8GB or more (for efficient performance, especially when handling multiple applications).

- **Storage:**

- At least 500MB of free disk space to store Python libraries, QR code image files, and project-related resources.

- **Monitor:**

- Standard resolution: 1280x720 pixels or higher (for better clarity and usability of the IDE).

1.2.2 SOFTWARE SPECIFICATION

The project is designed to be compatible with multiple operating systems and requires the following software components:

Operating System:

- The QR code generator can run on the following platforms:
 - Windows (Windows 10/11)
 - Linux (Ubuntu, Fedora, or other distributions)
 - macOS (Catalina, Big Sur, Monterey, or newer)

Python Version:

- The project is developed using Python 3.x.
 - Recommended version: Python 3.8 or later for optimal support and compatibility with the required libraries.

Integrated Development Environment (IDE):

- The following IDEs are recommended for coding, testing, and debugging the project:
 - PyCharm – Best suited for Python development with built-in package management.
 - VS Code – Lightweight and widely used with Python extensions.
 - Jupyter Notebook – Ideal for interactive coding and visualization.

Required Python Libraries:

To generate and customize QR codes, the following libraries must be installed:

- qrcode – The primary library for creating QR codes.
- PIL (Pillow) – Used for image processing and customization of QR codes.

Installation Command:

- `pip install qrcode[pil]`

This command ensures that all necessary dependencies are installed, allowing the seamless creation of QR codes in both ASCII format (for console display) and image format (for saving and sharing).

With the above system specifications, users can efficiently generate QR codes using Python. The project is designed to be lightweight and cross-platform, making it accessible on various hardware configurations while ensuring ease of development and execution.

1.3 SOFTWARE ENVIRONMENT

The project is developed in Python 3.x using the qrcode library for QR code generation. The Integrated Development Environment (IDE) used for development can be VS Code, PyCharm, or Jupyter Notebook. The libraries required for this project include:

- Programming Language: Python
- IDE: VS Code / PyCharm
- Library Used: qrcode (for QR generation), **PIL** (for image processing)

OVERVIEW OF THE SYSTEM ENVIRONMENT

The system environment for generating QR codes using Python refers to the complete setup of hardware and software that ensures the proper execution of the QR code generator application. In this project, the system environment must be configured correctly to enable smooth execution of Python code for QR code creation, image processing, and other associated tasks. This section details the required system components including the operating system, Python version, libraries, and necessary configurations.

OPERATING SYSTEM SETUP

The choice of operating system plays a crucial role in the development and execution of Python-based projects. QR code generation using Python can be done on various operating systems such as Windows, macOS, and Linux. Below is an overview of the operating systems commonly used in this environment:

WINDOWS SETUP

For Windows users, it is necessary to have a version of Windows 7 or later to run the required Python environment effectively. The following steps are recommended to set up the system:

Install Python: Visit the [official Python website](#) to download the latest version of Python 3. Ensure that during installation, the option to “Add Python to PATH” is checked to allow easy execution from the command prompt.

- Install Dependencies: Use the command prompt to install the required libraries by running
 - `pip install qrcode[pil] Pillow`
- IDE Setup: You can install PyCharm, Visual Studio Code, or any other preferred text editor to write and run the Python code. PyCharm and VS Code offer integrated tools for debugging and running scripts.

MACOS SETUP

macOS also supports Python-based QR code generation. The system comes with Python pre-installed, but you can install the latest version using the following steps:

- Install Python: Use Homebrew to install the latest Python version:
 - `brew install python`

- Install Libraries: Open the terminal and use pip to install required libraries:
 - `pip3 install qrcode[pil] Pillow`

INSTALLATION OF REQUIRED LIBRARIES

The main libraries required for QR code generation include qrcode and Pillow. These libraries allow for QR code creation, customization, and image handling.

QR CODE LIBRARY

The qrcode library is a simple and effective way to generate QR codes in Python. To install it, run the following command:

- `pip install qrcode[pil]`

SYSTEM STUDY

2.SYSTEM STUDY

The System Study is a critical phase of any project as it helps in understanding the existing system, identifying its limitations, and proposing solutions to overcome these limitations. In this section, we will analyze the Existing System and suggest a Proposed System that can be developed using Python for generating QR codes.

2.1 EXISTING SYSTEM

In this context, the Existing System refers to the traditional methods of generating QR codes and the challenges faced by users or developers while utilizing these systems. QR code generation has been prevalent for several years, but many existing methods are either complex, inefficient, or lack customization options.

1. OVERVIEW OF THE EXISTING SYSTEM

The existing QR code generation systems mostly involve using third-party software or web applications that allow users to input data and then generate QR codes. Some of the commonly used methods include:

- **Online QR Code Generators:** There are many online platforms that provide QR code generation services. Examples include websites like QR Code Generator, QRStuff, and GoQR.me. Users visit these websites, enter the data (such as a URL, text, or contact details), and generate a QR code that can then be downloaded.
- **Software Tools:** Some desktop software applications also exist to create QR codes. These tools are often limited in functionality and lack flexibility in customization. Popular tools include QR Code Studio and Zint Barcode Studio.

2. LIMITATIONS OF THE EXISTING SYSTEM

Although these existing systems are widely used, they present several challenges and limitations:

- **Lack of Customization:** Most online QR code generators offer very limited customization options in terms of color, size, or logo integration. Users may not be able to fully adjust the appearance of the generated QR codes.
- **Dependency on Internet Connectivity:** Online QR code generators require a stable internet connection to function. Users without reliable internet access are unable to use these tools effectively.
- **Limited Automation and Bulk Generation:** Existing tools generally work on a manual basis, which means that generating QR codes for large sets of data becomes tedious and time-consuming. Bulk generation or automated QR code creation is not supported by many existing systems.
- **Payment Systems:** Many payment platforms use QR codes to facilitate easy and secure transactions.

2.2 PROPOSED SYSTEM

The Proposed System suggests using Python programming to develop a more flexible, customizable, and automated QR code generation tool. This system will address the limitations of the existing methods while providing a more efficient, secure, and scalable solution for users.

1. OVERVIEW OF THE PROPOSED SYSTEM

The proposed system involves creating a Python-based solution for generating QR codes. It will utilize libraries such as qrcode and Pillow to provide an easy-to-use interface for generating and customizing QR codes. The system will be capable of:

- **Customizing QR Codes:** The user will have the ability to customize the color, size, border width, error correction level, and even add logos to the QR code. This feature makes the system highly flexible, enabling it to meet the specific branding or design requirements of businesses or individuals.
- **Batch Generation:** The proposed system will include an option to automate the process of generating QR codes for bulk data, which can be a significant time-saver for businesses, marketing teams, or developers needing to generate QR codes for large datasets (e.g., product inventory, event tickets, or promotional links).
- **Offline Functionality:** Unlike online tools, the proposed system will be able to run entirely offline once the Python environment and necessary libraries are set up. This eliminates dependency on internet connectivity, providing users with uninterrupted access to QR code generation capabilities.
- **Integration with Other Systems:** The proposed Python-based solution will allow easy integration into existing business systems. For example, businesses can automate the generation of QR codes linked to their inventory system or customer records.
- **Security and Privacy:** The proposed system will allow QR code generation to remain within the local machine, reducing concerns over uploading sensitive data to third-party servers. This will increase the privacy and security of the data being encoded.

2. FEATURES OF THE PROPOSED SYSTEM

- **Interactive User Interface:** An optional GUI (Graphical User Interface) can be created for non-technical users, enabling them to easily generate QR codes with the click of a button.
- **Data Validation and Error Handling:** The system will include built-in validation checks to ensure that the data being entered for QR code generation is in the correct format, minimizing errors and improving user experience.
- **Support for Multiple Data Types:** The proposed system will support the generation of QR codes for various types of data, including URLs, plain text, email addresses, contact information (vCards), and even geo-location data.

- **Customizable Error Correction Levels:** QR codes come with different error correction levels that allow them to still function even if parts of the code are damaged or obscured.

OBJECTIVE OF THE PROJECT

3. OBJECTIVE OF THE PROJECT

The primary objective of this project is to develop a QR code generator using Python that enables efficient and seamless sharing of digital information. By leveraging the qrcode library, the project aims to create a customizable and accessible QR code that can be scanned using any mobile device. The QR code generated in this project encodes the PPG College website URL, allowing users to access the site instantly by scanning the code.

This project serves as an example of how QR codes can be used to improve accessibility, enhance digital engagement, and facilitate information sharing efficiently.

The key objectives of this project include:

- **Generating and customizing QR codes using Python:**
 - Implementing the **qrcode** library to create unique and scannable QR codes.
 - Allowing modifications to the QR code (color, size, and data) to suit different requirements.
- **Providing a simple and effective solution for encoding URLs:**
 - The project demonstrates a practical approach to converting URLs into scannable QR codes.
 - It eliminates the need for manually typing web addresses, improving convenience.
- **Storing the QR code as an image file for easy distribution and sharing:**
 - Saving the QR code as a high-resolution image file (PNG format) makes it easier to print, share, or integrate into digital platforms.
- **Enabling console-based ASCII QR code visualization for instant access:**
 - The project also features an ASCII-based QR code display in the console, which provides an additional way to view the QR code, especially for quick testing.

- **Enhancing accessibility and engagement through QR code scanning:**
 - The QR code simplifies user access to the PPG College website, making it a useful tool for students, faculty, and visitors.
 - It demonstrates how QR codes can be effectively used in educational and institutional environments.
- **Showcasing a lightweight and efficient approach to QR code generation using Python:**
 - The project highlights how a few lines of Python code can generate a fully functional QR code, making it a valuable learning experience for students and developers interested in digital solutions.

By achieving these objectives, this project demonstrates the practical utility of QR codes in modern technology, particularly in education, marketing, and digital communication.

3.1 KEY FEATURES AND APPLICATIONS OF QR CODES IN THIS PROJECT:

- **Enhanced Accessibility:** QR codes provide an easy way for students, faculty, and visitors to access the college website without the need to manually type the URL.
- **Efficient Information Sharing:** The QR code can be printed on brochures, banners, or websites to quickly share essential details about the college.
- **Security and Authentication:** QR codes are widely used in authentication systems, such as student or staff identification, access control, and secure logins.
- **Marketing and Communication:** Educational institutions use QR codes in promotional materials, event invitations, and digital business cards for seamless engagement.

With the growing need for contactless and instant information retrieval, QR codes are becoming increasingly popular across industries. This project serves as an example of how QR codes can

be generated and customized using Python for practical real-world applications. It highlights the simplicity and effectiveness of Python-based automation in creating dynamic and interactive digital solutions.

FEATURES OF THE PROJECT

4.FEATURES OF THE PROJECT

This project provides a simple, efficient, and customizable way to generate QR codes using Python. It offers several key features that enhance its usability and functionality.

4.1 QR CODE GENERATION FROM ANY URL OR TEXT

- The project enables users to generate QR codes from any URL, text, or alphanumeric data.
- In this implementation, the QR code specifically encodes the URL of the PPG College website, allowing users to access it instantly by scanning the generated code.
- The system can be modified to encode other types of data, such as contact details, Wi-Fi credentials, or event information.

4.2 ASCII REPRESENTATION OF THE QR CODE IN THE CONSOLE

- The generated QR code is displayed directly in the console using ASCII characters.
- This feature is useful in environments where graphical interfaces are unavailable or when debugging the QR code output.
- The ASCII version provides a text-based representation of the QR code, making it possible to visualize without opening an image file.

4.3 SAVING THE QR CODE AS AN IMAGE FILE

- The project saves the generated QR code as a PNG image file (default filename: ppg.png).
- This makes it easy to share, print, or embed the QR code in documents, websites, or promotional materials.
- The saved image ensures that users can scan the QR code using any smartphone or QR scanner app.

4.4 CUSTOMIZABLE QR CODE COLORS AND SIZE

- The project supports customization of the QR code's foreground (fill) and background colors.
- Users can modify the color settings to make the QR code visually appealing and compatible with branding requirements.
- The QR code size can also be adjusted, ensuring compatibility with different display or print formats.

4.5 LIGHTWEIGHT AND EFFICIENT PYTHON IMPLEMENTATION

- The project is built using Python and the qrcode library, making it a lightweight solution that runs on most systems with minimal resource usage.
- Since the implementation requires only a few dependencies, it is easy to set up, deploy, and integrate into other applications.
- The QR code generation process is fast, making it ideal for real-time applications.

MODULE DESCRIPTION

5.MODULE DESCRIPTION

In this section, we will discuss the various modules involved in the QR Code generation project. Each module has a specific role in the process of generating, displaying, and saving QR codes, contributing to the overall functionality of the system.

5.1 QR CODE GENERATION MODULE

The QR Code Generation Module is the core component of the project. This module is responsible for converting the input data (such as text, URLs, or other information) into a QR code. It utilizes the qrcode library, which is a powerful Python library designed to create QR codes in a flexible and customizable manner.

Key Functions of the QR Code Generation Module:

- **Input Data Processing:** The user provides data in the form of a string, URL, or other textual data. This data is passed to the qrcode library to encode into a QR code.
- **QR Code Creation:** The qrcode library generates a matrix representing the data as a QR code. The module allows customization of the QR code's properties, such as:
 - ✧ **Error Correction Level:** This defines the amount of error correction added to the QR code. Higher error correction allows the QR code to be readable even if parts of it are obscured or damaged.
 - ✧ **Size and Border:** The module allows customization of the size of the QR code and the width of its borders.
 - ✧ **Color Customization:** Users can choose the foreground and background colors of the QR code to match branding or personal preferences.
 - ✧ **Customization:** The module also supports adding logos or images to the center of the QR code. This can be done using the **Pillow** library, which allows for image manipulation and placing logos in the QR code design.

Flow of the QR Code Generation Module:

1. **Receive Input Data** (URL, Text, etc.)
2. **Generate QR Code Matrix** with defined configurations.
3. **Customize the QR Code** (size, error correction, colors).
4. **Return the Generated QR Code** for further usage (display or saving).

5.2 CONSOLE DISPLAY MODULE

The Console Display Module is responsible for interacting with the user through the command-line interface (CLI). It provides feedback to the user about the status of QR code generation and allows for a simple input/output mechanism to control the program.

Key Functions of the Console Display Module:

- **User Input Handling:** The console allows the user to input the data they want to encode into the QR code. The module prompts the user to enter text, URLs, or other data via the command line.
- **Real-Time Feedback:** After the input is provided, the module provides real-time feedback to the user, indicating whether the QR code was successfully generated or if any errors occurred during the process.
- **Error Handling:** The console display module also handles errors such as invalid input, missing data, or issues with the QR code generation process. It informs the user with appropriate error messages.
- **User Interaction:** The module provides prompts that guide the user through the steps of the program. For example, it may ask the user for the desired file format or customization options (like color choices or error correction levels).

Flow of the Console Display Module:

1. **Prompt User for Input Data** (text, URL, etc.).
2. **Display Status Messages** (generation success, errors).
3. **Provide User-Friendly Error Handling** for invalid inputs.
4. **Show Success Message** once the QR code has been successfully generated.\

5.3 IMAGE SAVING MODULE

The **Image Saving Module** is responsible for saving the generated QR code image to the file system. This module provides the functionality to export the QR code in various formats, such as PNG, JPEG, or even SVG, based on the user's choice.

Key Functions of the Image Saving Module:

- **Save the QR Code Image:** Once the QR code is generated, this module saves the image to a specified location on the user's computer. It allows users to define the file name and choose the image format.
- **File Format Options:** The module supports multiple image formats, including:
 - **PNG:** A lossless format commonly used for QR code images due to its quality and transparency support.
 - **JPEG:** A lossy format suitable for smaller file sizes, though less commonly used for QR codes due to potential loss in quality.
 - **SVG:** A vector format that allows for infinite scalability, making it suitable for high-resolution displays and print.
- **Path Selection:** Users can specify the directory or folder where they want to save the generated QR code image. If no directory is specified, the image is saved to the current working directory.
- **Overwriting Protection:** If a file with the same name already exists in the target directory, the module prompts the user to either overwrite the existing file or save it with a different name to avoid data loss.

Flow of the Image Saving Module:

1. **Receive the Generated QR Code Image** from the QR Code Generation Module.
2. **Ask the User for the File Name and Format.**
3. **Save the Image** in the specified location and format.

4. **Confirm Save Operation** with the user.

PROJECT MANAGEMENT

6.PROJECT MANAGEMENT

Project management is a critical aspect of any software development process, as it helps organize, plan, and execute tasks efficiently. In this section, we will discuss various aspects of project management, including the planning, execution, resource allocation, timelines, risk management, and team collaboration for the QR code generation project using Python. The goal is to ensure that the project is delivered on time, within budget, and with all requirements met.

6.1 PROJECT PLANNING

Project Planning is the first and most important step in the project management process. It involves defining the scope of the project, setting objectives, and determining the resources and timelines needed to achieve the goals. For this QR code generation project, careful planning is required to ensure all modules function seamlessly, and the project is completed within the allotted time.

Project Objectives

- **Develop a QR Code Generation Tool:** The main objective of this project is to develop a tool that can generate QR codes from different data inputs (such as text, URLs, and contact information) using Python.
- **Customizable Features:** The project will allow users to customize the generated QR codes in terms of size, color, and error correction levels.
- **Image Saving Functionality:** The QR codes will be saved in various formats (such as PNG, JPEG, and SVG) to meet the needs of different users.
- **Ease of Use:** The project will provide a user-friendly interface, either through a command-line interface (CLI) or a graphical user interface (GUI), to interact with the system easily.

PROJECT SCOPE

The scope of the project includes:

- The development of a QR code generation system using Python.
- The creation of three main modules (QR Code Generation, Console Display, and Image Saving).
- Providing customization options for QR codes.
- Ensuring the tool can run on major operating systems like Windows, macOS, and Linux.
- Delivering a product that is both efficient and easy to use for end-users.

6.2 PROJECT EXECUTION

Project Execution refers to the implementation of the project plan, where actual development and testing take place. During the execution phase, the planned tasks are performed, resources are allocated, and the project team works towards the successful completion of the project.

Task Breakdown and Timeline

For the QR code generation project, the following tasks have been identified:

1. Requirement Analysis and Design (Week 1)

- Define user requirements.
- Create system architecture and design (modules and database structure).
- Prepare the initial prototype for the user interface (CLI/GUI).

2. Development Phase (Weeks 2-5)

- Develop the QR Code Generation module.
- Implement the Console Display module for user input and feedback.
- Develop the Image Saving module for saving the generated QR codes.
- Perform integration testing of the three modules.

3. Customization and Error Handling (Week 6)

- Add functionality for customizing QR code size, error correction levels, and color.

- Implement error handling for invalid input and other edge cases.

4. User Interface Design (Weeks 7-8)

- Develop a user-friendly CLI for input and feedback.
- If needed, create a simple GUI for non-technical users.

5. Testing and Debugging (Weeks 9-10)

- Perform unit tests for each module.
- Conduct system testing to ensure smooth integration between modules.
- Fix bugs and improve performance based on testing results.

6. Documentation and Final Review (Week 11)

- Prepare the final project documentation, including installation guides, user manuals, and technical documentation.
- Review the entire project to ensure all requirements are met.

7. Project Deployment and Delivery (Week 12)

- Deploy the tool to the end-users.
- Provide any necessary post-deployment support.

Tools and Technologies:

- **Python:** The programming language used for developing the QR code generator.
- **Libraries:** Libraries such as qrcode, Pillow, and Tkinter (for GUI if needed).
- **IDE/Editor:** Tools like PyCharm or Visual Studio Code for development.
- **Version Control:** Git will be used for code management and collaboration

6.3 RISK MANAGEMENT

Risk management is the process of identifying, assessing, and mitigating risks that could negatively affect the project's outcome. The following risks were identified for this QR code generation project:

1. Technical Challenges

- **Problem:** Incompatible versions of Python or libraries might cause issues in different environments (Windows, macOS, Linux).
- **Mitigation:** Test the application on all major operating systems early in the project. Ensure that the necessary Python versions and libraries are correctly installed on all platforms.

2. Time Constraints

- **Problem:** Tight deadlines may lead to rushed development and incomplete functionality.
- **Mitigation:** The project should be broken into manageable tasks with clear milestones and deadlines. Regular progress reviews should be held to track whether the project is on schedule.

3. Usability Issues

- **Problem:** The tool may not be user-friendly enough, especially for those who are not technically savvy.
- **Mitigation:** Focus on building an intuitive user interface. If time permits, develop a graphical user interface (GUI) in addition to the CLI for improved accessibility.

4. Integration Problems

- **Problem:** The modules (QR code generation, console display, and image saving) might not integrate well, causing system crashes or errors.
- **Mitigation:** Ensure thorough testing of each module before integration. Perform unit testing and integration testing to catch and fix bugs early.

5. External Dependencies

- **Problem:** External libraries, such as qrcode and Pillow, might not function as expected, especially if newer versions introduce breaking changes.
- **Mitigation:** Carefully track the versions of external libraries used, and ensure they are compatible with each other. Use version control systems like Git to maintain consistency across development environments.

6. Security Concerns

- **Problem:** The system could potentially be vulnerable to malicious inputs that could compromise the program or system security.
- **Mitigation:** Implement data validation and sanitize inputs to prevent issues such as code injection or corruption of the generated QR codes.

CONCLUSION

7. CONCLUSION

In conclusion, the QR code generation project using Python provides an efficient, flexible, and customizable solution for creating QR codes. Through the implementation of key modules, including the QR Code Generation Module, Console Display Module, and Image Saving Module, the system addresses the common challenges found in existing QR code generation methods. By offering features such as error correction, color customization, and bulk generation, the system enhances user experience and meets a wide range of needs, from personal to business use.

Additionally, the project's focus on ease of use, offline functionality, and cross-platform compatibility ensures its accessibility to a broad audience. The inclusion of a user-friendly interface, combined with seamless integration and image saving capabilities, makes the tool both practical and adaptable to different scenarios.

Ultimately, the successful completion of this project contributes to a deeper understanding of Python's capabilities in solving real-world problems, demonstrating how modular programming and careful project management can lead to the creation of effective and user-centered solutions. The QR code generator tool can be further enhanced by adding more features, such as batch generation, advanced security features, and API integration, making it a versatile tool for various industries and use cases.

BIBLIOGRAPHY

8. BIBLIOGRAPHY

REFERENCE BOOK

1. **Grayson, M. (2010).** *Python 3: The Complete Reference*. McGraw-Hill Education.
This book provides a comprehensive understanding of Python programming, covering all major aspects of the language used in the development of the QR code generation tool.
2. **Lutz, M. (2013).** *Learning Python (5th Edition)*. O'Reilly Media.
A widely regarded resource for mastering Python programming, this book covers fundamental concepts, libraries, and advanced topics, which were instrumental in the development of the project.
3. **Hernandez, J. (2015).** *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython*. O'Reilly Media.
While this book focuses on data analysis, the concepts introduced are relevant for handling data in Python applications, helping with understanding how to work with data when generating QR codes.
4. **Parker, D. (2017).** *Python Projects: 50+ Practical Projects to Help You Master Python*. Packt Publishing.
This book contains hands-on Python projects, including QR code generation, which helped in providing examples and practical knowledge in the implementation of the project.
5. **O'Reilly, T. (2014).** *QR Codes for Dummies*. Wiley Publishing, Inc.
This book offers a thorough introduction to QR codes, including how they are generated, their usage, and real-world applications. It was essential in understanding the theory behind QR code generation and its applications.

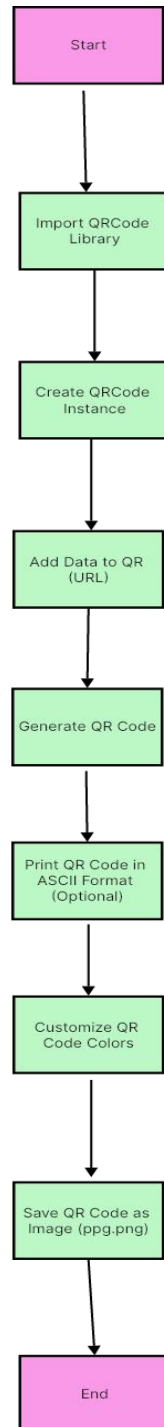
WEBSITE LINK

- <https://www.python.org/doc/>
- <https://pypi.org/project/qrcode/>
- <https://pillow.readthedocs.io/en/stable/>
- <https://stackoverflow.com/>
- https://www.w3schools.com/python/python_qrcode.asp

APPENDICES

9. APPENDICES

9.1 FLOW CHART



9.2 SOURCE CODE:

PPG CAS WEBSITE:

```
import qrcode # Install using: pip install qrcode[pil]

# Create a QR Code instance
qr = qrcode.QRCode()
qr.add_data("https://www.ppgarts.edu.in/") # Add URL
qr.make()

# Print QR code in ASCII format
qr.print_ascii()

# Save the QR code as an image with custom colors
img = qr.make_image(fill="yellow", back_color="white")
img.save("ppg.png")
```

PPG CAS INSTAGRAM WEBSITE:

```
import qrcode # Install using: pip install qrcode[pil]

# Create a QR Code instance
qr = qrcode.QRCode()
qr.add_data("https://www.instagram.com/ppgcascbe/") # Add URL
qr.make()

# Print QR code in ASCII format
qr.print_ascii()

# Save the QR code as an image with custom colors
img = qr.make_image(fill="yellow", back_color="white")
img.save("ppg.png")
```


PPG CAS FACEBOOK WEBSITE :

```
import qrcode # Install using: pip install qrcode[pil]

# Create a QR Code instance
qr = qrcode.QRCode()
qr.add_data("https://www.facebook.com/ppg.cas.3/?locale=en_GB/") #URL
qr.make()

# Print QR code in ASCII format
qr.print_ascii()

# Save the QR code as an image with custom colors
img = qr.make_image(fill="yellow", back_color="white")
img.save("ppg.png")
```

PPG CAS YOUTUBE WEBSITE :

```
import qrcode # Install using: pip install qrcode[pil]

# Create a QR Code instance
qr = qrcode.QRCode()
qr.add_data("https://www.youtube.com/@ppgcollegeinstitutions3773") #URL
qr.make()

# Print QR code in ASCII format
qr.print_ascii()

# Save the QR code as an image with custom colors
img = qr.make_image(fill="yellow", back_color="white")
img.save("ppg.png")
```

9.3 SCREENSHOT

PPG CAS QR CODE:



PPG CAS INSTAGRAM QR CODE:



PPG CAS FACEBOOK QR CODE:



PPG CAS YOUTUBE QR CODE:

